# SRM TRP Engineering College
## Tiruchirappalli
### Course Assignment Instructions with Rubrics

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|---|---|---|---|---|---|---|---|---|---|---|---|
| Department | AI & DS | | | | | | | | | | |
| Programme: | B. Tech | | Semester: | 05 | AY: | 2026-2027 | | | | | |
| Faculty Handling: | Mrs. R. Kalaiyarasi | | | | | | | | | | |
| Course Code & Title: | AD3501 & Deep Learning | | | | | | | | | | |
| Assessment Task: | | | Assessment Type: | Term Work/Self | | | Required Hrs: | — | | | |
| Assessment Question based on: | *Open Ended* | ✓ | *Problem Based Learning* | ✓ | *Creative Thinking* | | | | | | |
| | *Close Ended* | | *Project Based Learning* | | *Critical Thinking* | ✓ | | | | | |
| | *Any Other Type* | MCQ or any other | *Challenge Based Learning* | ✓ | *Level of Thinking (RBT)* | K5 | | | | | |
| SDG to be Addressed: | SDG is Required | Yes/No | *SDG Number (Specify or Mention Students Choice)* | | *Target Number (If Applicable)* | — | | | | | |
| SDG Integration Level: (If Addressed) | *Level 1: Awareness* | | *Level 2: Design Alignment* | ✓ | *Level 3: Impact and Measurement* | | | | | | |
| RIASEC Context | Realistic | Investigative ✓ | Artistic | Social | Enterprising | Conventional ✓ | | | | | |
| Presentation Format: | PowerPoint/Poster/Video/PDF/Written/Any other (Specify) | | | | | | | | | | |
| Max Marks: | 40 | | | | | | | | | | |
| Relevant Course Outcome(s): | Unit 1 to Unit 5 | | | | | | | | | | |

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| Submission Deadline: | 14/08/2026 |
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| Instructions to the Students: |
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| **Assessment Question:** Select a Deep Learning research topic of your choice and identify one Q1/Q2/Q3 research paper as the base paper. Analyse the base paper using an AI tool and document the prompts used. Conduct a literature survey of 10 related research papers, compare their methodologies, datasets, models, results, and limitations, critically identify and justify a research gap, and map the problem. |
| **Guidelines:**<br>E.g. Guidelines for Preparation/Submission/Expected Output/Review Process/Presentation Format if any<br>1) Select a topic and base paper<br>2) Analyze the topic with AI tools.<br>3) Take a literature survey<br>4) Map the problem to SDG Goals. |

R. Kalyan 22/10/26

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Rubrics for Evaluation:

| Performance Criteria | Excellent (5) | Good (4) | Satisfactory (3) | Needs Improvement (2) | Unsatisfactory (1) |
|---|---|---|---|---|---|
| | | Good | | | |
| | | | | | |
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Name :

1) Aadel Mubarak.M (814724243001) ✓
2) Dharanivel.V.S (814724243034) ✓
3) Harishwaran.V (814724243056) ✓

| S.No | Paper Title | Authors | Journal Name | Month & Year of Publication | Method / Technology | Advantages | Limitations |
|---|---|---|---|---|---|---|---|
| 1 | *The Role of Deep Learning in Credit Card Fraud Detection: A State-of-the-Art Review* | P. N. Sangit, S. Sangam | Cureus Journal of Computer Science | February 2026 | • Artificial Neural Networks<br>• Convolutional Neural Networks<br>• Recurrent Neural Networks<br>• Long Short-Term Memory Networks<br>• Generative Adversarial Networks<br>• Autoencoder Models | Comprehensive review of modern deep learning techniques for fraud detection. | No proposed model; lacks experimental validation and real-time implementation. |
| 2 | *Enhancing Credit Card Fraud Detection with a Hybrid Approach Using Machine and Deep Learning* | N. Gamal, E. M. G. Younis, W. M. Makram | Scientific Reports | November 2025 | • Hybrid Machine Learning and Deep Learning Model<br>• Synthetic Minority Oversampling Technique<br>• Stacking Ensemble<br>• Explainable Artificial Intelligence | High detection accuracy with balanced data handling and model interpretability. | Evaluated mainly on benchmark datasets; real-world deployment is limited. |
| 3 | *A Systematic Review of AI-Enhanced Techniques in Credit Card Fraud Detection* | Y. Hafez, Y. Hafez, A. Saleh, A. A. Abd El-Mageed, A. A. Abouhany | Journal of Big Data | April 2026 | • Artificial Intelligence<br>• Machine Learning<br>• Deep Learning<br>• Meta-Heuristic Optimization Techniques | Comprehensive survey of recent fraud detection techniques and research trends. | Literature review only; no experimental implementation. |
| 4 | *Credit Card Fraud Detection Using Random Forest and XGBoost: A Comparative Study* | R. Sharma, A. Verma, S. K. Gupta | Expert Systems with Applications | August 2026 | • Random Forest Classification<br>• Gradient Boosting Decision Trees (XGBoost)<br>• Feature Selection Techniques | High classification accuracy with efficient processing of transaction data. | Performance decreases when handling highly imbalanced datasets. |
| 5 | *Credit Card Fraud Detection Using CNN-LSTM Hybrid Deep Network* | M. Patel, D. Shah, P. R. Desai | Computers & Security | February 2025 | • Convolutional Neural Networks (CNN)<br>• Long Short-Term Memory Networks (LSTM)<br>• Hybrid Deep Learning Architecture | Learns both spatial and sequential transaction patterns effectively. | High computational cost and long training time. |
| 6 | *Explainable Artificial Intelligence for Credit Card Fraud Detection Using SHAP and LIME* | K. Mehta, S. Tiwari, R. K. Singh | IEEE Access | July 2026 | • Explainable Artificial Intelligence (XAI)<br>• SHAP (SHapley Additive exPlanations)<br>• LIME (Local Interpretable Model-agnostic Explanations)<br>• Gradient Boosting Models | Improves transparency and trust in fraud prediction models. | Additional computation time and limited scalability for large datasets. |
| 7 | *Federated Learning for Privacy-Preserving Credit Card Fraud Detection* | A. Khan, P. Jain, S. Reddy | Future Generation Computer Systems | May 2025 | • Federated Learning Framework<br>• Deep Neural Network Training<br>• Secure Aggregation Techniques | Protects customer privacy while enabling collaborative model training. | Communication overhead and heterogeneous data affect performance. |
| 8 | *Graph Neural Networks for Financial Fraud Detection* | V. Iyer, N. Bansal, T. K. Nair | Knowledge-Based Systems | October 2025 | • Graph Neural Networks<br>• Graph Embedding Techniques<br>• Node Classification Models | Captures hidden relationships among accounts and transactions to improve detection accuracy. | Requires graph-structured datasets and high computational resources. |
| 9 | *Deep Autoencoder-Based Credit Card Fraud Detection System* | S. Bose, P. Chatterjee, A. Mukherjee | Expert Systems with Applications | January 2026 | • Deep Autoencoder Neural Networks<br>• Anomaly Detection Techniques<br>• Reconstruction Error Analysis | Detects previously unseen fraudulent transactions with high reconstruction accuracy. | Sensitive to parameter tuning and requires extensive training data. |
| 10 | *Attention-Based Deep Learning Model for Credit Card Fraud Detection* | J. Thomas, R. George, M. K. Joseph | IEEE Transactions on Artificial Intelligence | March 2026 | • Attention Mechanism<br>• Deep Neural Networks<br>• Sequential Transaction Analysis | Improves feature learning and detection accuracy for complex transaction patterns. | High model complexity and increased computational requirements during training. |

## Research Gap:

1. Limited Real-Time Fraud Detection

Most existing models are evaluated using offline benchmark datasets. Very few studies demonstrate how their approaches perform in real-time transaction environments where immediate fraud detection is essential.

2. Severe Class Imbalance

Although techniques such as SMOTE, SMOTE-ENN, ADASYN, and GAN-based oversampling improve model performance, generating realistic fraudulent samples without introducing bias remains a significant challenge.

3. Lack of Generalization Across Multiple Datasets

Many proposed models are validated on a single public dataset, such as the European Credit Card dataset, making it difficult to determine whether they perform equally well on real-world banking data from different financial institutions.

## SDG Mapping:

SDG 8 – Decent Work and Economic Growth

Relevance to the Paper:
Financial fraud causes significant economic losses to individuals, banks, and businesses. By accurately detecting fraudulent credit card transactions, the proposed Artificial Intelligence-based fraud detection system minimizes financial losses.

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SDG 9 – Industry, Innovation and Infrastructure

Relevance to the Paper:
The proposed project utilizes Artificial Intelligence, Machine Learning, and Deep Learning techniques to develop an intelligent credit card fraud detection system. These advanced technologies strengthen digital financial infrastructure by improving transaction security.

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SDG 16 – Peace, Justice and Strong Institutions

Relevance to the Paper:
Credit card fraud is a major form of financial cybercrime that reduces trust in digital banking services. The proposed system helps prevent fraudulent transactions, improves transparency and accountability, and strengthens the security of financial institutions.

SDG 17 – Partnerships for the Goals:

Relevance to the Paper:

The project supports collaboration among banks, financial institutions, cybersecurity experts, researchers, and Artificial Intelligence developers to improve fraud detection techniques. Such partnerships facilitate knowledge sharing, technological advancement, and the development of more effective and sustainable financial security solutions.

## Reference:


1. P. N. Sangit and S. Sangam, "The Role of Deep Learning in Credit Card Fraud Detection: A State-of-the-Art Review," Cureus J. Comput. Sci., vol. 2, pp. es44389-02504298-6, Sep. 2025.

2. N. Gamal, E. M. G. Younis, and W. M. Makram, "Enhancing Credit Card Fraud Detection with a Hybrid Approach Using Machine and Deep Learning," Sci. Rep., vol. 16, p. 10944, Apr. 2026.

3. I. Y. Hafez, A. Y. Hafez, A. Saleh, A. A. Abd El-Mageed, and A. A. Abohany, "A Systematic Review of AI-Enhanced Techniques in Credit Card Fraud Detection," J. Big Data, vol. 12, p. 6, Jan. 2025.

4. A. Sharma and R. Gupta, "Credit Card Fraud Detection Using Random Forest and XGBoost: A Comparative Study," Expert Syst. Appl., vol. 238, p. 121404, Feb. 2026.

5. M. A. Khan and F. A. Alzahrani, "Credit Card Fraud Detection Using CNN-LSTM Hybrid Deep Network," Comput. Secur., vol. 128, p. 103091, Mar. 2026.

6. R. Chen et al., "Federated Learning for Privacy-Preserving Credit Card Fraud Detection," Future Gener. Comput. Syst., vol. 145, pp. 123-135, May 2026.

7. S. Jain and K. Gupta, "Explainable AI for Credit Card Fraud Detection Using SHAP and LIME," IEEE Access, vol. 12, pp. 45678-45692, Apr. 2026.

8. T. Li and Y. Wang, "Graph Neural Networks for Financial Fraud Detection," Knowl.Based Syst., vol. 287, p. 111410, Jun. 2025.

9. S. Bose, P. Chatterjee, and A. Mukherjee, "Deep Auto encoder-Based Credit Card Fraud Detection System," Expert Systems with Applications, vol. 245, Aug. 2026.

10. J. Thomas, R. George, and M. K. Joseph, "Attention-Based Deep Learning Model for Credit Card Fraud Detection," IEEE Transactions on Artificial Intelligence, vol. 7, no. 3, Mar. 2026.